

HDL-to-Netlist IC Design Flow

Digital Circuit Synthesis and Verification Using Synopsys Tools

Project Overview

This project focused on the implementation of the first steps of an integrated circuit design flow using hardware description language designs and netlist-level synthesis. The objective was to design digital circuits using HDL, synthesize them into gate-level netlists, and verify their functionality before and after synthesis. The project was developed as my undergraduate thesis in Electronics Engineering at Universidad del Valle de Guatemala.

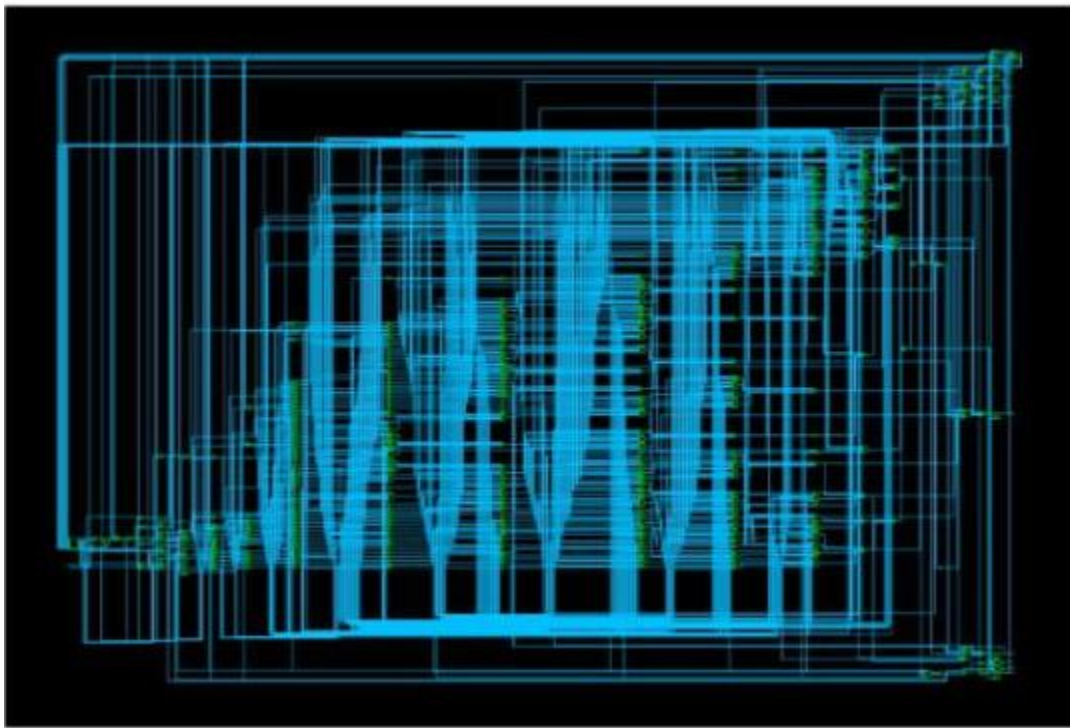


Figure 1. Circuit Schematic.

Motivation

Integrated circuit design requires a structured workflow that connects high-level circuit descriptions with lower-level gate implementations. Before a design can move toward physical implementation, it must be described, synthesized, verified, and checked against timing and design constraints.

System Concept

The design flow started with a digital circuit described in HDL. The circuit was then synthesized using Synopsys Design Vision and a 90 nm standard-cell library. The synthesis process generated a gate-level netlist representing the original digital circuit using logic gates from the selected technology library.

After synthesis, the behavior of the circuit was verified using simulation and formal equivalence checking. VCS was used to simulate the circuit before and after synthesis, while Formality was used to verify that the synthesized netlist preserved the functionality of the original HDL design.

My Role

My contributions to this project included:

- Installed and configured Synopsys tools for digital IC design and verification.
- Studied and used 90 nm CMOS standard-cell libraries for synthesis.
- Designed digital circuits using HDL.
- Synthesized HDL designs into gate-level netlists using Synopsys Design Vision.
- Applied design constraints to guide synthesis and improve implementation.
- Verified circuit behavior before and after synthesis using VCS.
- Used Formality to compare the HDL design and synthesized netlist.
- Documented the workflow and prepared educational material for future students.

Methodology

The methodology began with the preparation of the software environment, including Synopsys Design Vision, VCS, Formality, and the required CMOS technology libraries. After configuring the tools, digital test circuits were implemented using HDL and analyzed through the synthesis flow.

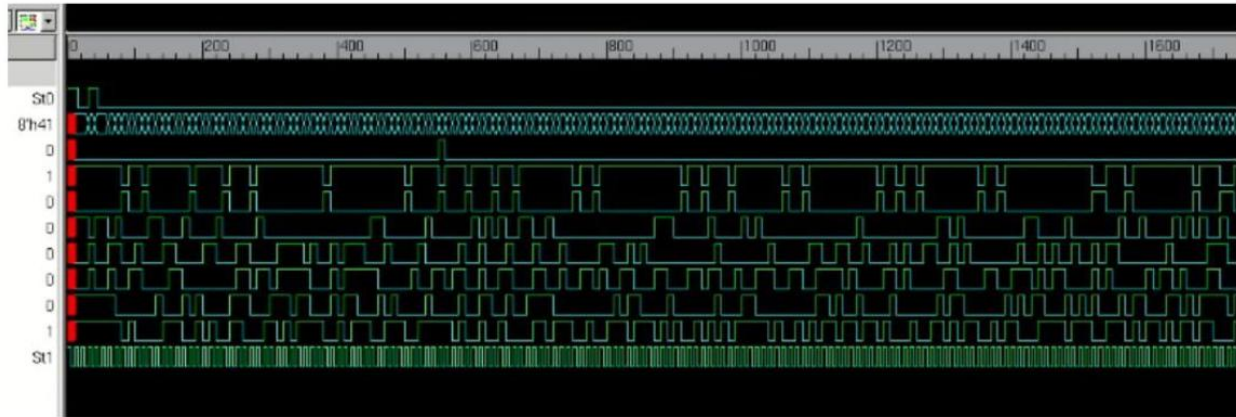


Figure 2. VCS verification.

Key Technical Areas

This project demonstrates experience in:

- Hardware description languages
- Digital integrated circuit design flow
- HDL-to-netlist synthesis
- Synopsys Design Vision, VCS and Formality
- CMOS standard-cell libraries

Results and Demonstrations

The project successfully established a basic HDL-to-netlist workflow for digital circuit design. Several digital circuits were synthesized and verified to understand the complete process from HDL description to gate-level representation.

The project also produced documentation and tutorial material intended to guide future students in the development of more complex nanometric CMOS circuit designs.

Engineering Relevance

This project is relevant to digital IC design, semiconductor engineering, and electronic design automation. It demonstrates familiarity with the early stages of ASIC design, including HDL description, synthesis, verification, and netlist generation.

Project Status

This project was completed as my undergraduate thesis in Electronics Engineering at Universidad del Valle de Guatemala. The work established a documented HDL-to-netlist synthesis and verification workflow using Synopsys tools, 90 nm CMOS libraries, and multiple digital circuit examples.